Week 21: Data Projects That Help Your Community

Find Data. Ask a Question. Support a Decision. Explain the Value.

Instructor / Mentor

Kiki Research Training Program

Week 21: Community Research Extension

Today's Mission: Be a Student Researcher

Imagine that your school's environmental club asks you to help the school or nearby residents make one better decision.

By the end of class, you can explain

1. what a real dataset contains and where it comes from;
2. which question it can answer—and which it cannot;
3. who could use the answer and what they could change; and
4. how to check whether that change actually helps.

Homework goal: Find one additional dataset yourself and propose one feasible community project.

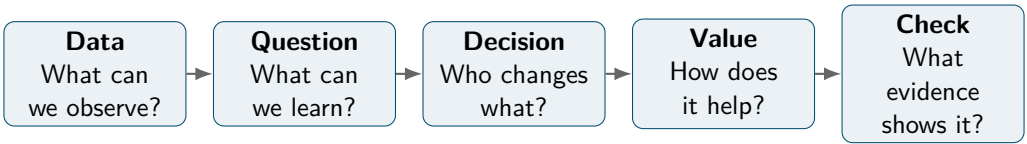
Our 120-Minute Workshop

Time	Focus	Your contribution
0–40 min	Scenarios and data: school energy and rooftop solar	Read sample records; calculate one possible benefit.
40–80 min	Evidence and decisions: peak demand and EV charging	Compare simple rules with AI; question the claimed value.
80–120 min	Project-design studio	Search for data, draft a project card, and give peer feedback.

No new advanced algorithm is required

A useful table, graph, or simple comparison can be a strong first research project. Coding and AI are tools, not the goal.

The Chain That Makes a Project Useful



Weak versus useful

"I will use AI on electricity data."

"I will compare similar school buildings to help the facilities manager choose one building for an after-hours equipment check."

How We Will Use the Assigned Report

- The report, dated **October 4, 2026**, contains four community-energy examples.
- Boston and Shenzhen are **example locations**. Adapt the reasoning to your community.
- Building, solar/weather, and market records are real data products.
- Lamp counts, panel size, rates, and vehicle behavior may be **scenario assumptions**.
- We summarize the report; we have not rerun its experiments.

Do not mix up three kinds of statement

Observed: a value in the source. **Estimated:** a calculation or prediction.
Assumed: a condition chosen for a scenario.

Source: assigned Community Energy AI Report, pp. 1–2, 18–19. Reported results, not new experiments.

Three Data Sources, Four Project Ideas

Source	What one row represents	Questions it helps explore
Boston BERDO	One annual building/property record: use, floor area, electricity	Which school merits a closer energy investigation?
NASA POWER	One location-day: solar radiation and weather	How much might solar panels generate? Can we estimate tomorrow's sunlight?
ISO New England	One load-zone hour: regional demand, weather, wholesale price	When might demand peak? How do charging schedules compare?

Remember: These sources do not directly contain your school's lamp schedule, roof condition, or individual drivers' needs.

Source: assigned Community Energy AI Report, pp. 3, 6, 10, 15, 18. Reported results, not new experiments.

Before Analyzing Anything, Read the Units

Quantity	Meaning	Example
Power: kW	How fast energy is being used	A 5 kW charger operates for 4 hours.
Energy: kWh	Total electricity used/generated	$5\ \text{kW} \times 4\ \text{h} = 20\ \text{kWh}$.
Intensity: kWh/m^2/year	Annual electricity per floor area	Compare large and small buildings more fairly.
Price: \$/MWh	Wholesale price per 1,000 kWh	Divide by 1,000 to obtain \$/kWh.

Quick check

Moving 20 kWh of charging from evening to night changes *when* energy is used. Does it necessarily reduce total energy? **No.**

Source: assigned Community Energy AI Report, pp. 2–3, 13–15. Reported results, not new experiments.

Example 1: Investigate School Energy

Project 1: Which School Should We Investigate First?

Community question

Among schools with reasonably comparable uses and sizes, which have unusually high annual electricity use per square meter?

- **User:** a school facilities manager, supported by an environmental club.
- **Data:** BERDO records published in 2024 about 2023 energy use.
- **Starting analysis:** filter school records, check floor-area units, calculate electricity intensity, and plot a comparison.
- **Decision:** choose a candidate for an equipment and operating-hours investigation.
- **Useful output:** a short investigation list with reasons and questions, not a public "wasteful schools" ranking.

Source: assigned Community Energy AI Report, pp. 3–5. Reported results, not new experiments.

Look at Three Real School Records

School address in the report	Area (m^2, rounded)	Annual electricity (kWh)
174 ipswich st	15,409	1,715,796
906 albany st	11,631	1,354,336
33-35 mildred av	15,979	1,803,120

Try the first row

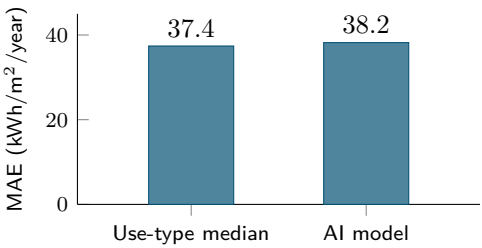
$$\text{Electricity intensity} = \frac{\text{annual electricity}}{\text{floor area}} \approx \frac{1,715,796}{15,409} = 111.4 \text{ kWh/m}^2\text{/year}.$$

Discuss: What could explain a high value besides waste? Electric heating, evening activities, or shared meters are possible explanations to investigate.

Source: assigned Community Energy AI Report, pp. 3–4. Reported results, not new experiments.

A More Complicated Model Is Not Automatically Better

The report checks predictions on **552 held-out records**, keeping properties separate between training and testing.



MAE: average absolute prediction error. Lower is better.

- Baseline: predict the median intensity for a building's use type.
- The AI model uses more building features.
- The simple baseline performs slightly better here.

Lesson: Start with a simple comparison. AI must earn its place.

Source: assigned Community Energy AI Report, pp. 4, 18–19. Reported results, not new experiments.

From a Candidate Building to a Possible Benefit

Illustrative assumptions, not measured equipment data: 100 lamps, each 40 W, run unnecessarily for 2 hours on 200 days per year.

$$\begin{aligned}
\text{Lamp power} &= 100 \times 40/1,000 = 4 \text{ kW},\\
\text{Avoided electricity} &= 4 \times 2 \times 200 = 1,600 \text{ kWh/year},\\
\text{Possible variable-charge savings} &= 1,600 \times \$0.25 = \$400/\text{year}.
\end{aligned}$$

What this calculation does not prove

It does not prove the school actually has these lamps or hours. A prediction residual is **not** a measured amount of avoidable electricity.

Source: assigned Community Energy AI Report, pp. 4–5. Reported results, not new experiments.

A Safe, Small School-Energy Pilot

1. Ask a staff mentor to help contact the facilities manager.
2. Confirm meter coverage, building use, and permission to inspect one space.
3. Record equipment power and unnecessary operating time without touching electrical equipment.
4. Let authorized staff implement any schedule change.
5. Compare before/after meter records for similar weather, occupancy, and school schedules.

Who benefits, and how do we know?

The school may save electricity and variable charges. Deliver a one-page check list; verify savings with actual records and the real tariff.

Source: assigned Community Energy AI Report, pp. 5, 17. Reported results, not new experiments.

Example 2: Explore Rooftop Solar

Project 2: Is Rooftop Solar Worth a Closer Look?

Community question

For a school or community center, how much electricity might a reference solar system generate, and what further information is needed before an installation decision?

- **Data:** NASA POWER daily solar radiation and weather at a chosen coordinate.
- **Report coverage:** 2020–2025; 2,192 days for each of Boston and Shenzhen.
- **A row:** location, date, solar radiation, temperature, humidity, and wind speed.
- **Useful first output:** an annual/monthly generation estimate plus an assumptions sheet.

Local missing information: usable roof area, shading, roof suitability, installation quotes, and electricity-use timing.

Source: assigned Community Energy AI Report, pp. 6–7, 18. Reported results, not new experiments.

Read the Solar Data and Make One Estimate

City	Date	Equivalent sun hours	Temperature (°C)
Boston	2025-06-14	2.033	14.4
Boston	2025-06-15	5.704	15.3
Shenzhen	2025-06-14	4.853	28.8
Shenzhen	2025-06-15	3.856	28.9

Solar radiation is recorded in kWh/m^2/day. Dividing by a reference irradiance of 1 kW/m^2 gives equivalent sun hours, **not daylight duration**.

Report's simplified, horizontal-panel scenario

$E \approx P_{\text{rated}} \times H_{\text{sun}} \times f_{\text{performance}}$

Assuming 100 kW and a 0.8 performance factor: $100 \times 3.8556 \times 0.8 \approx 308$ kWh/day.

Source: assigned Community Energy AI Report, pp. 6–7, 18. Reported results, not new experiments.

Generation Is Not the Same as Financial Value

The report's 100 kW reference system averages approximately **112,427 kWh/year in Boston** and **125,600 kWh/year in Shenzhen** over 2020–2025.

Boston scenario: state each assumption

If the school uses 60% of generated electricity on site, and each avoided purchased kWh saves \$0.25 in variable charges:

$$112,427 \times 0.60 \times \$0.25 \approx \$16,864/\text{year}.$$

- This is an estimated avoided purchase charge, **not net profit**.
- Export compensation, installation, maintenance, and financing still matter.
- **Decision supported:** ask for a roof assessment and quotes, not "install immediately."

Source: assigned Community Energy AI Report, pp. 6–7. Reported results, not new experiments.

Where Might AI Add Value to Solar?

Different question: Can past sunlight and weather predict tomorrow's total sunlight better than repeating yesterday?

2025 test error, reference-system kWh/day	Yesterday baseline	AI model
Boston, approximate MAE	109	92
Shenzhen, approximate MAE	92	84

Train on earlier years; select the model using 2024; test only after selection on 2025. Inputs must be available **before** the forecast is issued.

Better predictions are not yet proven community savings

Scheduling needs hourly generation, local demand, and deadlines. These errors compare NASA-based generation estimates, not measurements from an actual school installation.

Source: assigned Community Energy AI Report, pp. 8–9, 18–19. Reported results, not new experiments.

Example 3: Anticipate Busy Grid Hours

Project 3: When Might Many People Use Electricity?

Community question

Can a day-ahead forecast help a parking manager avoid charging during high-demand periods?

- **Source:** ISO New England SMD Hourly files, NEMA load zone, 2023–2025.
- **Fields:** `RT_Demand` (MW), `RT_LMP` ($/MWh), temperature (°F), local time.
- **Example:** January 1, 2025, 00:00: about 2,261 MW and 5.6°C after conversion.
- **Decision:** consider moving eligible charging away from forecast busy hours.

Match the scale of the data to the claim

Regional demand does not tell you your school's meter demand, bill, or a neighborhood circuit's capacity.

Source: assigned Community Energy AI Report, pp. 10, 18. Reported results, not new experiments.

A Fair Forecast Comparison

Item	Report's design
Information available	Earlier load/weather observations; target hour, day, season, holiday.
Forecast target	Each of tomorrow's 24 regional demand values.
Time separation	Train on 2023; choose settings on 2024; retrain on 2023–2024; test on 2025.
Simple baseline	Repeat last week's demand for the same hour.
Test-period MAE	Baseline approximately 294 MW; AI approximately 216 MW, over 8,712 test hours.

Do not leak the answer: tomorrow's observed temperature is not a valid input to a forecast made today. Published weather forecasts are a different, potentially available input.

Source: assigned Community Energy AI Report, pp. 10–11, 18. Reported results, not new experiments.

Average Accuracy Can Hide Important Failures

The report defines a high-demand day using a **research threshold** of about 3,713 MW, fixed from earlier-year daily peaks. It is not an official emergency limit.

	Actually high	Not high
Alert	14	15
No alert	22	—

$$\text{Recall} = \frac{14}{14 + 22} \approx 39\%.$$

$$\text{Precision} = \frac{14}{14 + 15} \approx 48\%.$$

- 36 high-demand days actually occurred in the report's test.
- The model missed 22 of them.
- The dash means the count is not reproduced here; these two metrics do not need it.
- This is a teaching/advisory result, not a reliable emergency-warning system.

Source: assigned Community Energy AI Report, pp. 11–12, 18. Reported results, not new experiments.

What Can a Parking Manager Actually Change?

Scenario: 10 vehicles each charge at 5 kW during a busy hour; all can charge later.

$$\text{Power moved out of that hour} = 10 \times 5 = 50 \text{ kW}.$$

- **Possible value:** avoid adding vehicle demand during a regional peak.
- **Service check:** vehicles receive the required energy before departure.
- **Local check:** ensure moving charging does not create a new parking-lot peak.
- **Cost check:** use the actual tariff; flat energy pricing may give no savings.

Say only what the evidence supports

Moving charging does not reduce total kWh. Construction-cost and emissions claims require additional evidence.

Source: assigned Community Energy AI Report, pp. 12, 17. Reported results, not new experiments.

Example 4: Compare EV Charging Schedules

Project 4: Charge on Arrival or Charge Overnight?

Community question

Which schedule meets drivers' needs while reducing peak-hour charging or cost under a specified tariff?

Assumption for one vehicle	Consequence
Arrives 18:00; leaves 07:00	Thirteen possible one-hour charging slots.
Needs 20 kWh from the grid	Every strategy must deliver exactly the same energy.
Charger power 5 kW	Select four full one-hour slots.
Charging may pause	The four slots need not be consecutive.

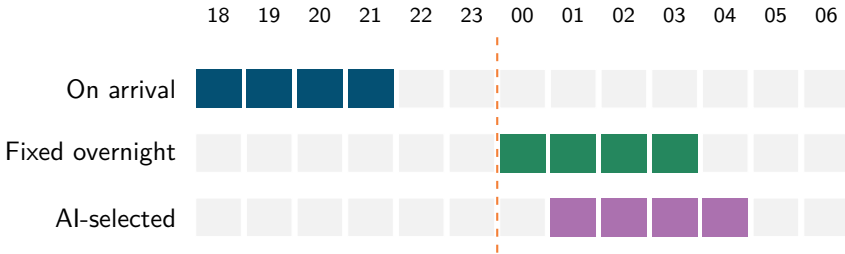
Observed inputs: historical regional demand and wholesale prices.

Not observed: a real school's fleet, arrival times, and required energy.

Source: assigned Community Energy AI Report, pp. 13–15. Reported results, not new experiments.

Three Schedules, the Same 20 kWh

Example arrival: June 24, 2025, at 18:00. Each filled square represents 1 hour at 5 kW.



AI-selected hours use **predicted regional demand**, not tomorrow's actual demand or actual prices. Historical prices are used afterward to evaluate cost.

Source: assigned Community Energy AI Report, pp. 13, 15. Reported results, not new experiments.

Compare AI with All Relevant Simple Rules

Report scenario: 1,000 identical vehicles, 360 valid nights in 2025, and 7.2 million kWh delivered by *every* strategy.

Strategy	Wholesale-price replay cost	Less than arrival charging
Charge on arrival	$603,417	$0
Fixed 00:00–04:00	$412,591	$190,826
AI-selected low-load hours	$401,197	$202,220
Last-week low-load hours	$399,550	$203,867

The important comparison

AI improves on fixed overnight charging by $11,394 in this replay, but the last-week rule costs $1,647 less than AI. AI does not win every comparison.

Source: assigned Community Energy AI Report, pp. 14–16. Reported results, not new experiments.

What Is the Value—and Who Receives It?

- **Scheduling value:** charging can move out of regional high-demand hours; verify the parking lot's own peak separately.
- **Cost value:** the report replays wholesale prices. A household pays according to its retail contract, not necessarily those prices.
- **Flat-rate counterexample:** at \$0.25/kWh, all four strategies have the same energy charge: 7.2 million $\times$ \$0.25 = \$1.8 million.
- **Who benefits:** a purchasing organization only if its contract reflects the cheaper hours; drivers only if benefits reach them.
- **Missing checks:** real driver deadlines, charger limits, local peaks, and implementation costs.

Do not overclaim

Total electricity saved in this scenario: **0 kWh**. Emissions reduction was not established because hourly emissions data were not provided.

Source: assigned Community Energy AI Report, pp. 14–17. Reported results, not new experiments.

Your Community Project Studio

What the Four Examples Have in Common

Project	Data-supported answer	User's decision	Community value to verify
School energy	Candidate buildings or unusual intensity	Inspect one space and its operating hours	Metered kWh and actual charges avoided
Rooftop solar	Generation estimate under stated conditions	Seek roof assessment and quotes	On-site generation used and net financial value
Peak forecast	Predicted busy hours and missed alerts	Coordinate flexible charging	Demand moved, deadlines met, false alerts tracked
EV schedules	Fair comparison of feasible schedules	Choose a timer or scheduling rule	Actual-contract cost and local peak, with equal energy

A project needs a next action

A graph is evidence. A prediction is an estimate. Neither is automatically a community benefit.

Source: assigned Community Energy AI Report, pp. 17. Reported results, not new experiments.

Audit a Dataset Before Committing to It

1. **Publisher and access:** Is it an identifiable primary source? Can you actually view or download records?
2. **Meaning:** What does one row represent? Which fields and units matter?
3. **Coverage:** Which locations and dates are represented? What is missing?
4. **Resolution:** Annual, daily, or hourly? A region, a station, or one building?
5. **Availability:** Was the information available at the time of any proposed prediction?
6. **Permission:** What reuse terms, privacy restrictions, or school approvals apply?

Keep a small dataset card

Record the dataset title, publisher, exact link, access date, selected location/period, units, and three limitations. Save five real sample rows.

Turn a Broad Topic into a Testable Question

Too broad

"How can AI solve energy problems in my city?"

Bounded and useful

"For one permitted school parking lot, would fixed overnight charging reduce variable electricity charges relative to charging on arrival, while meeting every driver's departure deadline?"

Question template

For **[one place/user]** during **[a defined period]**, can **[an analysis or change]** improve **[a measurable outcome]** compared with **[a simple baseline]**, while respecting **[a constraint]**?

A descriptive question is also valid: "Which public records suggest that one school deserves a closer investigation?" No AI model is required.

Choose the Simplest Analysis That Can Answer It

Question type	First method	Useful check
What is happening?	Summary table, counts, trend plot	Units, missing values, coverage
How do options compare?	Compare scenarios with equal service	Same energy, dates, and constraints
What might happen next?	Persistence or seasonal baseline first	Later-period error, missed events
Did a change help?	Comparable before/after observations	Weather, occupancy, other changes

If you choose AI

Explain what it adds, compare with a simple baseline, and use only inputs available at decision time. A weaker AI result is still a valid research finding.

A Small Reproducible Workflow

1. Save the source reference and keep an unchanged copy of permitted raw data.
2. Choose one manageable location and time window.
3. Check duplicates, missing values, units, and timestamp conventions.
4. Calculate one summary or comparison and create one clearly labeled plot.
5. Check the result against a simple baseline or a hand calculation.
6. Write the decision, assumptions, limitations, and next validation step.

For time data, a row is not always a normal hour

Local daylight-saving transitions can produce 23- or 25-hour days. Do not silently drop or double-count them when summing energy or evaluating schedules.

Time-handling example: assigned report, p. 18.

Describe Value Without Inventing Results

Unsupported claim	Defensible proposal statement
"My project will save $10,000."	"I will estimate savings under the actual tariff and validate a permitted pilot."
"High use means the school is wasteful."	"High use makes it a candidate; equipment and usage patterns may explain it."
"My AI is useful because it is accurate."	"I will test whether its advice improves a decision compared with a simple rule."

Useful value statement

"**[User]** could use **[output]** to change **[action]**. I would measure **[outcome]** against **[baseline]**, while checking **[important constraint]**."

A New Data Source Can Lead to a New Project

These are **search directions**, not a dataset already chosen for your homework.

New source to explore	Possible student question	Potential user and output
NOAA daily weather records	Which months had more hot days in a chosen past period?	School club: a seasonal preparation chart; not a same-day safety forecast.
EPA historical air-quality records	How often did available local records show elevated pollution during a chosen season?	School staff: a historical pattern summary; not medical advice or a live alert.
Your city's open-data catalog	Which locations have fewer documented shade trees or transit options?	Community group: a map of candidates for further investigation.

Check first: local records may be missing, delayed, or too coarse. Historical weather or pollution records do not establish current conditions at your school. Sources: NOAA CDO; EPA AirData.

Studio: Find Data, Then Draft a Project Card

1. **15 minutes: data search.** Choose a local issue; search an official agency or local open-data catalog. Open a specific dataset, not just a search result.
2. **15 minutes: project card.** Write a question, selected fields, simple analysis, user, decision, possible value, and limitations.
3. **10 minutes: peer/mentor check.** Ask whether the available data actually support the question and whether the user could act on the result.

If a link or dataset is unusable

Record what failed and switch sources. Do not invent sample records or claim access you do not have. Class work is a draft; finalize your independent choice for homework.

Your One-Project Proposal Card

Part	What to write
Community and question	One place, one user, one bounded question.
Data and access	A verified additional dataset; relevant fields, units, dates, and five sample rows.
Analysis and comparison	Three to five steps; a simple baseline; one planned figure/table.
Decision and value	Who could change what; how the result might help.
Validation and limits	One success metric, one practical constraint, two limitations, and a fallback.
First step	A realistic action you can take within seven days, with permission if needed.

Scale it down: one dataset, one local area, one main question, and one deliverable is enough for a first proposal.

Peer Check: Can the Project Really Work?

Ask your partner or mentor:

1. Can you show actual rows from the source and explain each important unit?
2. Does the data's location and time resolution match your question?
3. Is the analysis possible with tools you already understand?
4. Who might use the output, and what concrete action could they take?
5. What simple comparison would show that your method adds value?
6. What evidence would make you reduce or reject your claim?

Responsible research

Do not publish identifiable bills or student/driver routines. Do not operate electrical equipment. Any real intervention belongs to authorized adults.

Homework: One New Dataset, One Independent Project

Required task

Find **one additional dataset not used in the report** and propose **one project of your own** using the examples' data–question–decision–value structure.

- **Dataset card:** title, publisher, exact source link, access date, coverage, row meaning, field units, and reuse/access conditions.
- **Evidence:** five real sample rows and a saved extract or documented query/screenshot.
- **Proposal:** 1–2 pages with one question, a simple analysis/baseline, a community user, expected value, and validation plan.
- **Independence:** choose the dataset and draft the question yourself before any optional AI feedback. Disclose that feedback.

No complete model, finished experiment, or new code is required. Use `week21_homework.tex` for instructions and the planning worksheet.

Exit Ticket: Explain Your Project in 60 Seconds

1. "My community user is _____."
2. "My additional dataset contains _____."
3. "My question is _____."
4. "I will first compare _____."
5. "The answer could help the user decide _____."
6. "I will check usefulness by measuring _____."

The takeaway

Start with a question you can answer and a person who can use the answer. Clear evidence and honest limits matter more than a complicated model.

Source and Discovery Guide

Where to Find the Worked Examples

Assigned local report: `prism-uploads/Community_Energy_AI_Report.pdf`.

Pages	Read for
1–2	Purpose, units, and meaning of inputs and predictions
3–5	School-building comparison and lamp-schedule scenario
6–9	Solar estimation, forecast evaluation, and missing local evidence
10–12	Regional demand, forecast errors, and missed alerts
13–16	Equal-energy charging schedules and wholesale-price replay
17–19	Community deliverables, data provenance, and methods

The report mentions a reproduction package, but only the PDF is supplied here. Its numerical results are attributed to the report, not independently reproduced in this lesson.

Official Entry Points for the Report's Data

1. Boston BERDO: Data Disclosure. Follow "Search Prior Data Disclosures"; for this example select the 2024 disclosure of 2023 energy records.
   Report's catalog: BERDO dataset catalog.
2. NASA POWER: Daily API documentation. Select location, dates, solar radiation, and weather parameters. Check the response's units and missing-data conventions.
3. ISO New England: Zonal Information / SMD Hourly Data. Select year and NEMA sheet; check local times and original field units.

Reproducibility habit

Record the specific resource/version and observation years. A newer portal release may not match the exact historical file used in the report.

Official entry pages checked October 4, 2026; resource coverage and access can change.

Discovery Links for Your Additional Dataset

- NOAA Climate Data Online: search Daily Summaries by station/location and past date range. Inspect which observations are actually present.
- EPA AirData daily download: choose pollutant, year, geography, and monitor. Keep location and pollutant units with the extract.
- **Local discovery:** search "[your city or county] open data [trees / transit / energy / waste]" and identify the original publisher and downloadable records.

A portal is not a dataset

Submit the *specific* dataset or query you opened. A new URL or another year of the same BERDO, NASA POWER, or ISO-NE dataset is not an additional dataset for this homework.

You may choose another official source. Availability in your own area must be checked; these links do not guarantee suitable local coverage. Discovery entry pages checked October 4, 2026.